UF UNIVERSITY of FLORIDA

Department of Mathematics

MAP 2302
Exam 4A
Spring 2023

A. Sign your bubble sheet on the back at the bottom in ink.

B. In pencil, write and encode in the spaces as indicated below. Expect -5 for errors.

1) Name (last name, first initial, middle initial)

2) UF ID number

3) Section number

C. Under "special codes" code in the test ID numbers 4, 1.

1	2	3	●	5	6	7	8	9	0
●	2	3	4	5	6	7	8	9	0

D. At the top right of your answer sheet, for "Test Form Code", encode A.

● B C D E

E. 1) This test consists of 20 multiple choice questions of five points in value.

2) The time allowed is 120 minutes.

3) You may write on the test.

4) Raise your hand if you need more scratch paper or if you have a problem with your test. DO NOT LEAVE YOUR SEAT UNLESS YOU ARE FINISHED WITH THE TEST.

F. KEEP YOUR BUBBLE SHEET COVERED AT ALL TIMES.

G. When you are finished:

1) Before turning in your test **check carefully for transcribing errors**. Any mistakes you leave in are there to stay.

2) You must turn in your scantron to your exam proctor. Be prepared to show your picture I.D. with a legible signature.

3) The answers will be posted in Canvas shortly after the exam.

NOTE: Be sure to bubble the answers to questions 1−20 on your scantron.

Questions 1 − 20 are worth 5 points each.

1. Find the minimum radius of convergence of a series solution about $x_0 = 1$ for

$$(7x^2 + 6x - 1)y'' - (x^2 - 7)y' + xy = 0.$$

a. $\sqrt{7}$ b. $\frac{6}{7}$ c. $\sqrt{\frac{6}{7}}$

d. 3 e. 6

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2. Determine the Taylor polynomial of degree 3 for the solution to $y' = \frac{1}{x+y+1}$, $y(0) = 0$.

a. $y = x - 2x^2 + \frac{5}{3}x^3$

b. $y = x - x^2 - \frac{5}{3}x^3$

c. $y = 1 + x - x^2 + \frac{5}{3}x^3$

d. $y = x + 2x^2 + \frac{5}{3}x^3$

e. $y = x - x^2 + \frac{5}{3}x^3$

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3. Find the recurrence relation for a series solution to $y'' - x^2y' - xy = 0$ about $x_0 = 0$.

a. $a_{n+2} = \dfrac{na_{n-1}}{(2n+1)(n+1)}$

b. $a_{n+2} = \dfrac{na_n}{(n+2)(n+1)}$

c. $a_{n+2} = \dfrac{na_{n-1}}{(n+2)(n+1)}$

d. $a_{n+2} = \dfrac{na_{n-1}}{(n+2)(2n+1)}$

e. $a_{n+2} = \dfrac{na_n}{(n+2)(2n+1)}$

4. $\mathscr{L}^{-1}\left\{\dfrac{s}{s^2-14s+58}\right\} =$

a. $e^{-7t}\cos(3t) + \frac{7}{3}e^{-7t}\sin(3t)$

b. $e^{7t}\cos(3t) + \frac{2}{3}e^{7t}\sin(3t)$

c. $e^{7t}\cos(3t) - \frac{7}{3}e^{7t}\sin(3t)$

d. $e^{7t}\cos(3t) + \frac{7}{3}e^{7t}\sin(3t)$

e. $e^{7t}\cos(2t) - \frac{7}{3}e^{7t}\sin(2t)$

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5. Find $y_2(x)$ such that $y_1(x)$ and $y_2(x)$ are linearly independent solutions to

$$x^2y'' + 2xy' - 6y = 0, \quad y_1 = x^2,\ x > 0.$$

a. $y_2(x) = \ln(x^{-3})$

b. $y_2(x) = x^{-3}$

c. $y_2(x) = x^{-1}$

d. $y_2(x) = x^{-2}$

e. $y_2(x) = e^x x^{-3}$

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6. Solve $x^2y'' + xy' - y = \ln x$ by Variation of Parameters. Assume $x > 0$.

a. $\frac{c_1}{x} + c_2x - x\ln x$

b. $\frac{c_1}{x} + c_2x - \ln x$

c. $\frac{c_1}{x} + c_2x - e^x$

d. $\frac{c_1}{x} + c_2x^{-2} - \ln x$

e. $\frac{c_1}{x} + c_2x^2 - \ln x$

7. Find the form of a particular solution to $y'' + 3y' + 5y = x^4e^{5x}$. Do not solve for the coefficients. Assume a-d, h, and k are constants.

a. $y_p = (ax^4 + bx^3 + cx^2 + hx + k)e^{5x}$

b. $y_p = x(ax^3 + bx^2 + cx + d)e^{5x}$

c. $y_p = x^2(ax^3 + bx^2 + cx + d)e^{5x}$

d. $y_p = (ax^5 + bx^4 + cx^3 + dx^2 + hx + k)e^{5x}$

e. $y_p = ax^4e^{5x}$

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8. Find $\mathscr{L}^{-1}\left\{\frac{1}{s-s^2}\right\}$.

a. $(t-1)e^t$

b. $t^2 \cos t$

c. $e^t - t$

d. $1 - e^t$

e. 1

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9. Solve the IVP

$$ty' + 2y = t^2 - t + 1 \text{ and } y(1) = \tfrac{1}{2}.$$

a. $y(t) = \frac{1}{4}t^2 - \frac{1}{3}t + \frac{1}{2} + \frac{1}{12t^2}$

b. $y(t) = \frac{1}{4}t^2 + \frac{1}{3}t - \frac{1}{2} + \frac{1}{12t^2}$

c. $y(t) = \frac{1}{4}t^2 + \frac{1}{3}t - \frac{1}{2} - \frac{1}{12t^2}$

d. $y(t) = \frac{1}{4}t^2 - \frac{1}{3}t + \frac{1}{4} - \frac{1}{12t^2}$

e. $y(t) = \frac{1}{4}t^2 + \frac{1}{3}t + \frac{5}{8} + \frac{1}{12t^2}$

10. Let $y' = x^2 + y^2$ with $y(0) = 2$ and a step size $h = 0.1$. Use Euler's method to calculate y_2. Note: $x_0 = 0,\ y_0 = y(0) = 2$.

a. $y_2 = 2.846$

b. $y_2 = 2.901$

c. $y_2 = 2.968$

d. $y_2 = 2.977$

e. $y_2 = 2.914$

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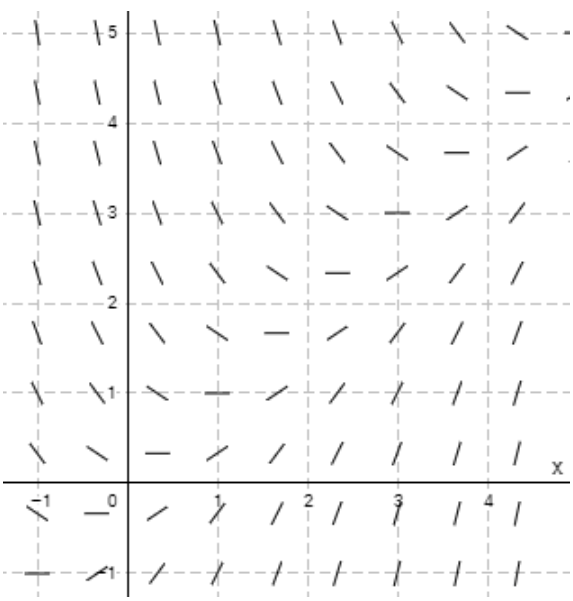
11. To which differential equation does the direction field below correspond?



a. $y' = x - y$

b. $y' = x + y$

c. $y' = x \sin y$

d. $y' = x^2 + y$

e. $y' = \ln y$

12. The differential equation $y' = \dfrac{xe^y}{y}$ is

a. linear

b. homogeneous

c. separable

d. exact

e. Bernoulli

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13. Find the interval of convergence for $\displaystyle\sum_{n=1}^{\infty} \frac{19}{n^2+8n}(x-7)^n$.

a. $(-\infty, \infty)$

b. $[6, 8)$

c. $(6, 8)$

d. $(6, 8]$

e. $[6, 8]$

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14. Express $\displaystyle\sum_{n=9}^{\infty} \frac{a_n}{n+4} x^{n+4}$ with generic term x^k.

a. $\displaystyle\sum_{k=9}^{\infty} \frac{a_{k-4}}{k} x^k$

b. $\displaystyle\sum_{k=13}^{\infty} \frac{a_{k+4}}{k} x^k$

c. $\displaystyle\sum_{k=13}^{\infty} \frac{a_{k-4}}{k} x^k$

d. $\displaystyle\sum_{k=13}^{\infty} \frac{a_{k+4}}{k} x^{k+4}$

e. $\displaystyle\sum_{k=9}^{\infty} \frac{a_k}{k} x^{k-4}$

15. Find the first four nonzero terms of a power series expansion about $x = 0$ for a general solution to the DE

$$y' + (x+2)y = 0 \ .$$

a. $y(x) = a_0(1 - 2x - 3x^2/2 - x^3/3 + ...)$

b. $y(x) = a_0(1 - 2x + 3x^2/2 - x^3/3 + ...)$

c. $y(x) = a_0(1 - 2x + 3x^2/4 - x^3/3 + ...)$

d. $y(x) = a_0(1 - 2x + 3x^2/2 - x^3/6 + ...)$

e. $y(x) = a_0(1 - 2x + 3x^2/2 + x^3/3 + ...)$

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16. Determine all of the singular points for the DE

$$(x^2 - 1)y'' + (1 - x)y' + (x^2 - 2x + 1)y = 0.$$

a. $x = -1, 1$

b. $x = -1, 0, 1$

c. $x = -1$ only

d. $x = 0$ only

e. There are no singular points.

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17. Find the first four nonzero terms of a power series expansion of the solution to the initial value problem.

$$x' + (\sin t)x = 0, x(0) = 1$$

a. $1+(1/2)t^2+(1/6)t^4-(31/720)t^6+...$

b. $1-(1/2)t^2+(1/3)t^4-(31/720)t^6+...$

c. $1-(1/2)t^2+(1/6)t^4-(31/720)t^6+...$

d. $1-(1/2)t^2-(2/3)t^4-(31/720)t^6+...$

e. $1 - (1/2)t^2 + (1/6)t^4 - (31/360)t^6 + ...$

18. Find the first four nonzero terms of a power series expansion of the solution to the DE about the given value for t_0.

$$x' + 2(t-1)x = 0,\ t_0 = 1$$

a. $a_0[1 - (t-1)^2 + (1/4)(t-1)^4 - (1/6)(t-1)^6 + ...]$

b. $a_0[1 - (t-1)^2 + (1/2)(t-1)^4 + (1/12)(t-1)^6 + ...]$

c. $a_0[1 - (t-1)^2 + (1/3)(t-1)^4 - (1/6)(t-1)^6 + ...]$

d. $a_0[1 - (t-1)^2 + (1/2)(t-1)^4 - (1/6)(t-1)^6 + ...]$

e. $a_0[1 + (t-1)^2 + (1/2)(t-1)^4 + (1/6)(t-1)^6 + ...]$

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19. Solve the exact DE.

$$2xy - 9x^2 + (2y + x^2 + 1)\,\tfrac{dy}{dx} = 0$$

a. $y^2 + (x^2+1)\,y - 3x^3 = C$

b. $y^2 + (x^2+1)\,y - 2x^3 = C$

c. $y^2 + (x^2+1)\,y - x^3 = C$

d. $y^2 + (x^2-1)\,y - 3x^3 = C$

e. $y^2 + (x^2-1)\,y + 3x^3 = C$

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20. Evaluate the integral $\displaystyle\int_{-\infty}^{\infty} 2^{-t}\delta(t+1)dt$.

a. $\frac{1}{2}$
b. 0
c. 1
d. -2
e. 2

Elementary Laplace Transforms

$f(t)$	$\mathcal{L}\{f(t)\}$
C	$\frac{C}{s}, s > 0$
t^n	$\frac{n!}{s^{n+1}}, s > 0$
e^{at}	$\frac{1}{s-a}, \quad s > a$
$e^{at}t^n$	$\frac{n!}{(s-a)^{n+1}}, \quad s > a,\ n = 0, 1, 2, \ldots$
$\sin(bt)$	$\frac{b}{s^2+b^2}, s > 0$
$\cos(bt)$	$\frac{s}{s^2+b^2}, s > 0$
$e^{at}\sin(bt)$	$\frac{b}{(s-a)^2+b^2}, \quad s > a$
$e^{at}\cos(bt)$	$\frac{s-a}{(s-a)^2+b^2}, \quad s > a$
$\sinh(bt)$	$\frac{b}{s^2-b^2}, s > \lvert b\rvert$
$\cosh(bt)$	$\frac{s}{s^2-b^2}, s > \lvert b\rvert$
$u(t-a)$	$\frac{e^{-as}}{s}, s > 0$
$\delta(t-a)$	e^{-as}

Properties of the Laplace transform

$\mathcal{L}\{f+g\} = \mathcal{L}\{f\} + \mathcal{L}\{g\}$

$\mathcal{L}\{cf\} = c\mathcal{L}\{f\}$ for any constant c

$\mathcal{L}\{e^{at}f(t)\}(s) = F(s-a)$ where $F(s) = \mathcal{L}\{f\}(s)$

$\mathcal{L}\{f'\}(s) = sF(s) - f(0)$

$\mathcal{L}\{f''\}(s) = s^2F(s) - sf(0) - f'(0)$

$\mathcal{L}\{f^{(n)}\}(s) = s^nF(s) - s^{n-1}f(0) - s^{n-2}f'(0) - \ldots - f^{(n-1)}(0)$

$\mathcal{L}\{t^nf(t)\} = (-1)^n\frac{d^nF}{ds^n}(s)$

$\mathcal{L}\{ty'\}(s) = -sY'(s) - Y(s)$ where $Y(s) = \mathcal{L}\{y\}(s)$

$\mathcal{L}\{ty''\}(s) = -s^2Y'(s) - 2sY(s) + y(0)$